

Appendix A

Mathematical Background

A.1. DETERMINANTS

A *determinant* is a square array of numbers or symbols, called *elements*, with a rule for finding the value of the array. Determinants are denoted in several ways, for example,

$$\det(a) = |a_{ij}|$$

$$= \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} \quad (\text{A.1})$$

the element a_{ij} being in the i th row and j th column. The determinant obtained by deleting the i th row and j th column is the *minor*, M_{ij} , of the element a_{ij} . The quantity, $(-1)^{i+j}M_{ij}$, is the *cofactor* of a_{ij} , A_{ij} .

One rule for finding the value of $\det(a)$ can be stated thus (Pipes and Harvill, 1978, pp. 85-6):

$$\left. \begin{aligned} \sum_j a_{ij}A_{ij} &= \sum_i a_{ij}A_{ij} = \det(a) \\ \sum_j a_{ij}A_{kj} &= 0 \quad i \neq k \\ \sum_i a_{ij}A_{ik} &= 0 \quad j \neq k \end{aligned} \right\} \quad (\text{A.2})$$

This equation states that the value of $\det(a)$ can be found by summing the products of each element of any row (or any column) and its cofactor. However, if we sum products of any row (or any column) by the cofactors of a different row (column), the sum is zero.

We illustrate Equation (A.2) by finding the value of the determinant

$$\begin{vmatrix} 2 & 4 \\ 3 & 5 \end{vmatrix}$$

Using the first row, the value is $(2 \times 5 - 4 \times 3) = -2$; using the first column, we have $(2 \times 5 - 3 \times 4)$

$= -2$. Determinants are usually evaluated using the first row or column, but the rule is better illustrated by expanding the following determinant using the second column:

$$\begin{vmatrix} 4 & 2 & 7 \\ 3 & -8 & 8 \\ 9 & 6 & 5 \end{vmatrix} = (-1)^{1+2}(2) \begin{vmatrix} 3 & 8 \\ 9 & 5 \end{vmatrix} \\ + (-1)^{2+2}(-8) \begin{vmatrix} 4 & 7 \\ 9 & 5 \end{vmatrix} \\ + (-1)^{3+2}(6) \begin{vmatrix} 4 & 7 \\ 3 & 8 \end{vmatrix} \\ = -2(-57) - 8(-43) - 6(11) \\ = 392.$$

The same result is obtained regardless of which row or column is used.

If we expand the preceding determinant using the third row and cofactors of the first row, we obtain

$$(-1)^{3+1}(9) \begin{vmatrix} -8 & 8 \\ 6 & 5 \end{vmatrix} + (-1)^{3+2}(6) \begin{vmatrix} 3 & 8 \\ 9 & 5 \end{vmatrix} \\ + (-1)^{3+3}(5) \begin{vmatrix} 3 & -8 \\ 9 & 6 \end{vmatrix} = 0$$

This illustrates the second result in Equation (A.2).

The solution of a set of simultaneous linear equations can be expressed in terms of determinants using *Cramér's rule* (Pipes and Harvill, 1970, p. 101). Given a set of n equations in the n unknowns $x_1, x_2, \dots, x_n$,

$$\left. \begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= k_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= k_2 \\ \vdots & \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= k_n \end{aligned} \right\} \quad (\text{A.3a})$$

the solution is

$$x_j = \det(a_j) / \det(a) \quad (\text{A.3b})$$

where $\det(a)$ is the determinant of the coefficients a_{ij} , and $\det(a_j)$ is $\det(a)$ with the j th column replaced by the column $(k_1, k_2, \dots, k_n)$. As an example, we solve the equations

$$\begin{aligned}4x_1 + 2x_2 + 7x_3 &= -4 \\3x_1 - 8x_2 + 8x_3 &= -15 \\9x_1 + 6x_2 + 5x_3 &= 11\end{aligned}$$

The value of $\det(a)$ is found to be 392. Then,

$$\begin{aligned}\det(a_1) &= \begin{vmatrix} -4 & 2 & 7 \\ -15 & -8 & 8 \\ 11 & 6 & 5 \end{vmatrix} \\&= -4 \begin{vmatrix} -8 & 8 \\ 6 & 5 \end{vmatrix} + 15 \begin{vmatrix} 2 & 7 \\ 6 & 5 \end{vmatrix} + 11 \begin{vmatrix} 2 & 7 \\ -8 & 8 \end{vmatrix} \\&= 664\end{aligned}$$

so that $x_1 = 664/392 = 1.69$. The values of x_2 and x_3 are found by substituting the k column for the second and third columns of $\det(a)$.

A.2. MATRICES

A *matrix* is a rectangular array of numbers or symbols (elements) plus rules for their manipulation. Matrices are represented in several ways; for example, a matrix of order $(m \times n)$ may be written

$$\mathcal{A} = \|a_{ij}\| = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{vmatrix} \quad (\text{A.4a})$$

Matrices of the same order $(m \times n)$ can be added (or subtracted) by adding (subtracting) corresponding elements; thus, if $\mathcal{A}, \mathcal{B}, \mathcal{C}$ are matrices of the same order, then $\mathcal{C} = \mathcal{A} \pm \mathcal{B}$ has elements $c_{ij} = a_{ij} \pm b_{ij}$. To form the product $\mathcal{A}\mathcal{B}$, the orders of $\mathcal{A}$ and $\mathcal{B}$ must be $(m \times p)$ and $(p \times n)$, the product $\mathcal{C}$ being of order $(m \times n)$. The elements c_{ij} are given by the formula

$$c_{ij} = \sum_k a_{ik} b_{kj} \quad (\text{A.4b})$$

that is, c_{ij} is the sum of the products, element by element, of the i th row of $\mathcal{A}$ and the j th column of $\mathcal{B}$. In general, $\mathcal{A}\mathcal{B} \neq \mathcal{B}\mathcal{A}$. To illustrate,

$$\begin{aligned}&\begin{vmatrix} 2 & -4 & 1 \\ 3 & 5 & -6 \end{vmatrix} \begin{vmatrix} 6 & 2 & -8 & 1 \\ 5 & -3 & 9 & 0 \\ 2 & -7 & 3 & 2 \end{vmatrix} \\&\quad (2 \times 3) \quad \quad (3 \times 4) \\&= \begin{vmatrix} -6 & 9 & -49 & 4 \\ 31 & 33 & 3 & -9 \end{vmatrix} \\&\quad (2 \times 4)\end{aligned}$$

The element c_{13} is, for example, $2 \times (-8) + (-4) \times 9 + 1 \times 3 = -49$.

The *transpose* of a matrix, $\mathcal{A}^T$, is the matrix $\mathcal{A}$ with rows and columns interchanged. A matrix of order $(n \times 1)$ is a *column matrix*, for example, the k column in Equation (A.3a). A matrix of order $(1 \times n)$ is a *row matrix*. A *null matrix* $\mathbf{0}$ has zeros for all elements; thus, if $\mathcal{A} = \mathcal{B}$, then $\mathcal{A} - \mathcal{B} = \mathbf{0}$ and each element of $\mathcal{A}$ equals the corresponding element of $\mathcal{B}$.

An $(n \times n)$ matrix is a *square matrix* and has a determinant, $\det(a)$, whose elements are those of the matrix; elements a_{ii} form the *principal diagonal* of the matrix. A square matrix in which all elements off the principal diagonal are zero but at least one of the diagonal elements is not zero is a *diagonal matrix*. The *identity (unit) matrix* is a diagonal matrix with elements $a_{ii} = 1$. A *symmetric matrix* is a square matrix with $a_{ij} = a_{ji}$; if $a_{ij} = -a_{ji}$, the matrix is *skew symmetric*.

If each element of a square matrix is replaced by its cofactor in $\det(a)$ of the matrix and the result transposed, we get the *adjoint* of $\mathcal{A}$, $\text{adj } \mathcal{A}$. The *reciprocal* of $\mathcal{A}$ is

$$\mathcal{A}^{-1} = \{1/\det(a)\} \text{adj } \mathcal{A} \quad (\text{A.5a})$$

[provided that $\det(a) \neq 0$]; this follows from Equation (A.2) because on using Equation (A.4b) to find the elements of the product $\mathcal{A}\mathcal{A}^{-1}$, all off-diagonal terms are zero whereas all diagonal terms equal $\det(a)$.

The equivalent of division by a square matrix is multiplication by $\mathcal{A}^{-1}$. If $\mathcal{A}$ is not square, we make it square by forming the product $\mathcal{A}^T \mathcal{A}$, which is always square. Thus, the solutions of the matrix equations $\mathcal{A} = \mathcal{A}\mathcal{N}$ and $\mathcal{P} = \mathcal{P}\mathcal{A}$, where $\mathcal{A}$ and $\mathcal{P}$ are, respectively, square and nonsquare are

$$\mathcal{N} = \mathcal{A}^{-1} \mathcal{A} \quad \mathcal{P} = (\mathcal{P}^T \mathcal{P})^{-1} \mathcal{P}^T \mathcal{P} \quad (\text{A.5b})$$

Matrices can be used to solve equations. Equation (A.3a) in matrix form is

$$\mathcal{A}\mathcal{X} = \mathcal{K} \quad (\text{A.6a})$$

where $\mathcal{A}$ is a square matrix of the coefficients, $\mathcal{X}$ is a column matrix of the unknowns x_i , and $\mathcal{K}$ is a column matrix of the k_i 's. The solution is

$$\mathcal{X} = \mathcal{A}^{-1} \mathcal{K} \quad (\text{A.6b})$$

Equating the $(n \times 1)$ column matrix on the left with the product, which is also a $(n \times 1)$ column matrix, we get the n values of x_i .

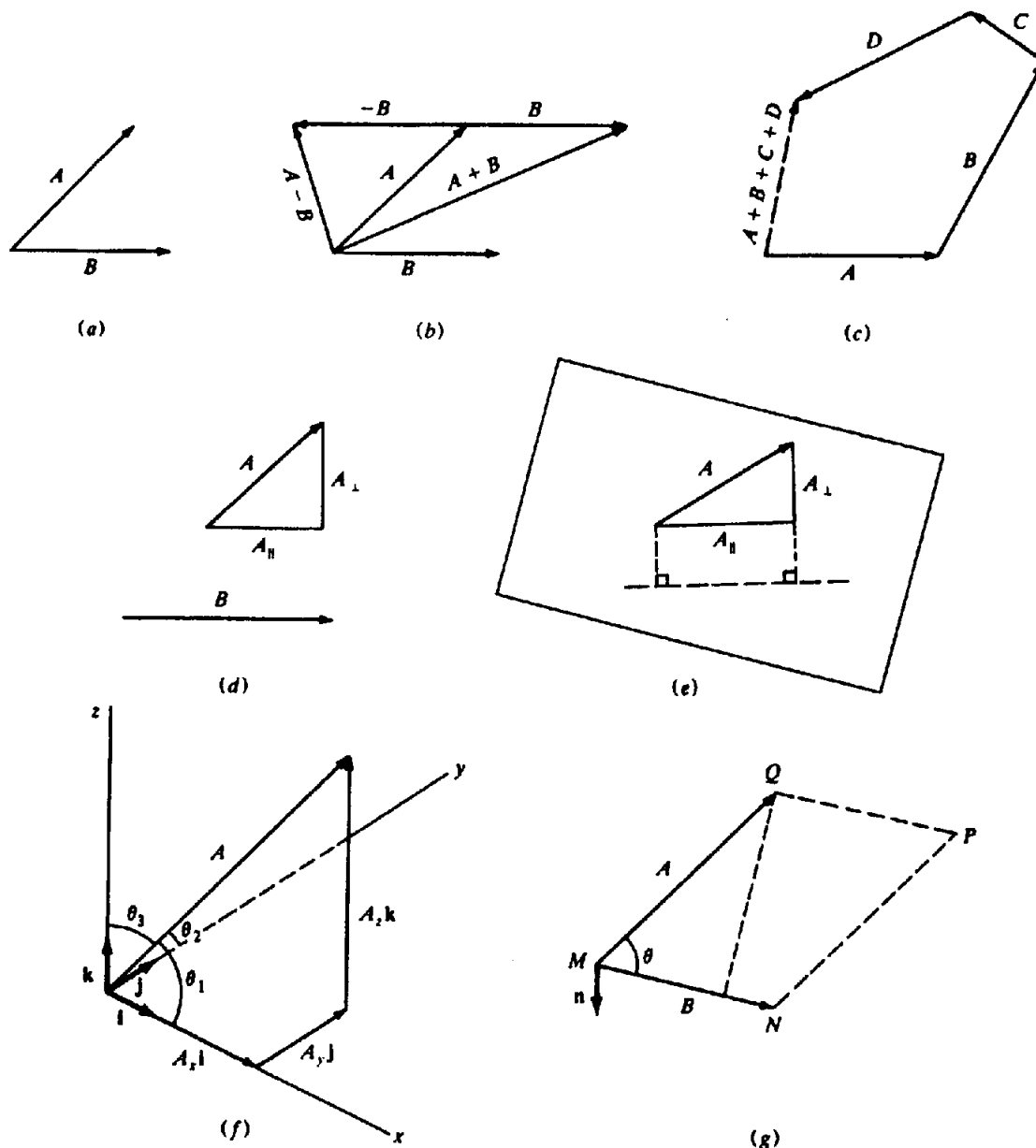


Figure A.1. Various operations on vectors. (a) Two vectors to be added or subtracted. (b) Addition and subtraction of vectors in (a). (c) Summing several vectors. (d) Resolving A into components parallel to and perpendicular to B . (e) Resolving A into components parallel to and perpendicular to a plane. (f) Resolving A into rectangular components. (g) Illustrating the products $A \cdot B$ and $A \times B$.

A.3. VECTOR ANALYSIS

A.3.1. Basic Theory

All measurable quantities can be divided into two classes: (i) *scalars*, which have magnitude only, hence can be completely specified by a single number, for example, temperature and density, and (ii) *vectors*, which have both magnitude and direction so that they are specified by giving a number (magnitude) and a direction, for example, force and velocity. Vectors will be denoted by boldface letters.

A vector can be represented by an arrow whose length is proportional to the magnitude and whose direction is that of the vector. To add two vectors, A and B (Fig. A.1a), we move one of them as in Figure A.1b, and the sum is the vector from the initial to the final point. The vector $-B$ is B reversed in direction, so we subtract B from A by adding $-B$ to A (Fig. A.1b). Several vectors can be added together as in Figure A.1c.

Calculations involving vectors are often facilitated by resolving them into *components*. Any vector can be regarded as the sum of several arbitrary

vectors; for example, in Figure A.1c the sum ($A + B + C + D$) can be a given vector with arbitrary components A , B , C , and D . Sometimes we resolve a vector into components parallel to and perpendicular to another vector (Fig. A.1d) or to a plane (Fig. A.1e). More often we use components parallel to the axes of a coordinate system; for example, in Figure A.1f we take vectors of unit length, i , j , and k , along the x , y , and z axes, and the components of A are the scalar quantities a_x , a_y , a_z . Then

$$A = a_x i + a_y j + a_z k \quad (\text{A.7})$$

Also, from Figure A.1f, we get

$$a_x/A = \cos \theta_1 = \ell$$

$$a_y/A = \cos \theta_2 = m$$

$$a_z/A = \cos \theta_3 = n$$

where θ_1 , θ_2 , and θ_3 are the angles between A and the three axes, (ℓ, m, n) are the *direction cosines* of A , and A is the *magnitude* of A . Because $A^2 = a_x^2 + a_y^2 + a_z^2$, on dividing by A^2 , we have

$$\ell^2 + m^2 + n^2 = 1 \quad (\text{A.8})$$

We can add and subtract vectors by adding and subtracting components:

$$\begin{aligned} \text{if} \quad A &= 2i + 3j - 4k \\ B &= 4i - j - 5k \\ C &= 2i - 3j + 3k \end{aligned}$$

then

$$\begin{aligned} A - B + C &= (2 - 4 + 2)i + (3 + 1 - 3)j \\ &\quad + (-4 + 5 + 3)k \\ &= j + 4k \end{aligned}$$

A.3.2. Vector Products

Products involving vectors are of three basic types. A vector can be multiplied by a scalar to give a new vector changed only in magnitude; this property is used to write a vector A in the form Aa_1 where $A = |A|$ = magnitude of A and a_1 is a unit vector parallel to A . The other two basic products involve two vectors and are written $A \cdot B$ and $A \times B$. The first, called the *dot product* or *scalar product*, is equal to the scalar quantity $(AB \cos \theta)$ where θ is the acute angle between A and B (Fig. A.1g); it is also equal to the magnitude of one of the vectors times the component of the other parallel to the first. The second product, $A \times B$, called the *cross product* or *vector product*, is defined as the vector $(AB \sin \theta)n$, where n is a unit vector perpendicular to the plane containing A and B and in the direction of advance

of a right-handed screw rotated from A to B (Fig. A.1g).

The definition of the dot product shows that $A \cdot B = B \cdot A$, also

$$i \cdot i = j \cdot j = k \cdot k = 1$$

$$i \cdot j = j \cdot k = k \cdot i = 0$$

$$\begin{aligned} A \cdot B &= (a_x i + a_y j + a_z k) \cdot (b_x i + b_y j + b_z k) \\ &= (a_x b_x + a_y b_y + a_z b_z) \end{aligned} \quad (\text{A.9})$$

$$A \cdot A = A^2 = a_x^2 + a_y^2 + a_z^2 \quad (\text{A.10})$$

From the definition, we see that $A \times B = -B \times A$, also (for a right-handed set of orthogonal coordinate axes)

$$i \times j = k \quad j \times k = i \quad k \times i = j$$

$$i \times i = j \times j = k \times k = 0$$

$$\begin{aligned} A \times B &= (a_x i + a_y j + a_z k) \times (b_x i + b_y j + b_z k) \\ &= (a_y b_z - a_z b_y)i + (a_z b_x - a_x b_z)j \\ &\quad + (a_x b_y - a_y b_x)k \end{aligned} \quad (\text{A.11a})$$

$$= \begin{vmatrix} i & j & k \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} \quad (\text{A.11b})$$

Figure A.1g shows that

$$|A \times B| = \frac{1}{2}(\text{area of parallelogram } MNPQ)$$

This means that a plane area can be considered as a vector with direction along the normal to the plane. Therefore we can resolve areas into components; in particular, an infinitesimal area ds can be written

$$ds = (dy dz)i + (dz dx)j + (dx dy)k \quad (\text{A.12})$$

Vector products may involve more than two vectors. Because $(B \cdot C)$ is a scalar, it can only enter into a triple product in the form $(B \cdot C)A$. On the other hand, $(B \times C)$ is a vector, so we can have the products $A \cdot (B \times C)$ and $A \times (B \times C)$. From Equations (A.9) and (A.11b) we find that

$$A \cdot (B \times C) = B \cdot (C \times A) = C \cdot (A \times B) \quad (\text{A.13a})$$

$$\begin{aligned} &= -A \cdot (C \times B) = -B \cdot (A \times C) \\ &= -C \cdot (B \times A) \end{aligned} \quad (\text{A.13b})$$

$$= \begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix} \quad (\text{A.13c})$$

Thus, interchanging dot and cross does not change the value, but changing the cyclic order changes the sign. The triple product $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$ can be expanded twice using Equation (A.11b), the result being

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C} \quad (\text{A.14})$$

Although any number of vectors can be multiplied together, products of more than three are rare.

A.3.3. The Vector Operator ∇

If $\phi(x, y, z)$ is the value of a scalar function at $P(x, y, z)$, then the value at a nearby point $Q(x + dx, y + dy, z + dz)$ is $\phi + d\phi$ where

$$d\phi = (\partial\phi/\partial x) dx + (\partial\phi/\partial y) dy + (\partial\phi/\partial z) dz \quad (\text{A.15})$$

Writing $d\mathbf{r}$ for the vector displacement from P to Q , we have

$$d\mathbf{r} = dx \mathbf{i} + dy \mathbf{j} + dz \mathbf{k}$$

and Equation (A.15) can be written as

$$d\phi = (\nabla\phi) \cdot d\mathbf{r} \quad (\text{A.16})$$

where

$$\nabla\phi = (\partial\phi/\partial x)\mathbf{i} + (\partial\phi/\partial y)\mathbf{j} + (\partial\phi/\partial z)\mathbf{k} \quad (\text{A.17a})$$

$$= \text{gradient of } \phi(x, y, z) \quad (\text{A.17b})$$

$$= \text{grad } \phi \quad (\text{A.17c})$$

Dividing both sides of Equations (A.15) and (A.16) by $d\mathbf{r}$ gives

$$d\phi/d\mathbf{r} = (\nabla\phi) \cdot \{ (dx/dr)\mathbf{i} + (dy/dr)\mathbf{j} + (dz/dr)\mathbf{k} \} \quad (\text{A.18a})$$

$$= (\nabla\phi) \cdot (\ell\mathbf{i} + m\mathbf{j} + n\mathbf{k}) \quad (\text{A.18b})$$

$$= (\nabla\phi) \cdot \mathbf{r}_1 \quad (\text{A.18c})$$

$\mathbf{r}_1$ being a unit vector along PQ with direction cosines (ℓ, m, n) . Because $(\nabla\phi) \cdot \mathbf{r}_1$ is the derivative of $\phi(x, y, z)$ at P as we go in the direction $\mathbf{r}_1$, it is called the *directional derivative*.

The vector $\nabla\phi$ is fixed at a given point but $(\nabla\phi) \cdot \mathbf{r}_1$ varies as the direction $\mathbf{r}_1$ is varied. When $\mathbf{r}_1$ is parallel to $\nabla\phi$, the product will have its maximum value because $|\mathbf{r}_1| = 1$ and $\cos \theta$ is a maximum for $\theta = 0$. Therefore $\nabla\phi$ is a vector in the direction of, and equal to, the maximum rate of increase of $\phi(x, y, z)$.

The expression $\{(\partial/\partial x)\mathbf{i} + (\partial/\partial y)\mathbf{j} + (\partial/\partial z)\mathbf{k}\}$ is known as a *vector operator*; it is represented by the symbol ∇ (pronounced del). When we apply it to a scalar function $\phi(x, y, z)$, we get $\nabla\phi$ or $\text{grad } \phi$. It can also be applied to a vector function of position, $\mathbf{A}(x, y, z)$, by taking either the dot or the cross product. In the first case we get

$$\begin{aligned} \nabla \cdot \mathbf{A} &= \{ (\partial/\partial x)\mathbf{i} + (\partial/\partial y)\mathbf{j} + (\partial/\partial z)\mathbf{k} \} \\ &\quad \cdot (A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k}) \\ &= (\partial A_x/\partial x) + (\partial A_y/\partial y) + (\partial A_z/\partial z) \end{aligned} \quad (\text{A.19a})$$

$$= \text{divergence of } \mathbf{A}(x, y, z) \quad (\text{A.19b})$$

$$= \text{div } \mathbf{A}(x, y, z) \quad (\text{A.19c})$$

In the second case, Equation (A.11b) gives

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A_x & A_y & A_z \end{vmatrix} \quad (\text{A.20a})$$

$$\begin{aligned} &= (\partial A_z/\partial y - \partial A_y/\partial z)\mathbf{i} \\ &\quad + (\partial A_x/\partial z - \partial A_z/\partial x)\mathbf{j} \\ &\quad + (\partial A_y/\partial x - \partial A_x/\partial y)\mathbf{k} \end{aligned} \quad (\text{A.20b})$$

$$= \text{curl } \mathbf{A} \quad (\text{A.20c})$$

The operator ∇ involves differentiation, and when it is applied to the product $\phi\mathbf{A}$, ϕ being a scalar function, we get two terms involving differentiation of ϕ and $\mathbf{A}$ separately. Therefore,

$$\nabla \cdot (\phi\mathbf{A}) = \nabla\phi \cdot \mathbf{A} + \phi\nabla \cdot \mathbf{A} \quad (\text{A.21})$$

$$\nabla \times (\phi\mathbf{A}) = \nabla\phi \times \mathbf{A} + \phi\nabla \times \mathbf{A} \quad (\text{A.22})$$

Treating ∇ as a true vector, we would have

$$\begin{aligned} \nabla \cdot \nabla &= \nabla^2 \\ &= \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2 \\ &= \text{Laplacian} \end{aligned} \quad (\text{A.23})$$

$$\nabla \times \nabla\phi = 0 \quad \nabla \cdot (\nabla \times \mathbf{A}) = 0 \quad (\text{A.24})$$

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} \quad (\text{A.25})$$

Equations (A.21) to (A.25) can be verified by substituting the definition of ∇ . [Note that Equation (A.25) is valid only in rectangular cartesian coordinates.]

A.3.4. Vector Theorems

The divergence of a vector has a geometrical interpretation that is often very useful. Let $\mathbf{A}(x, y, z)$ be

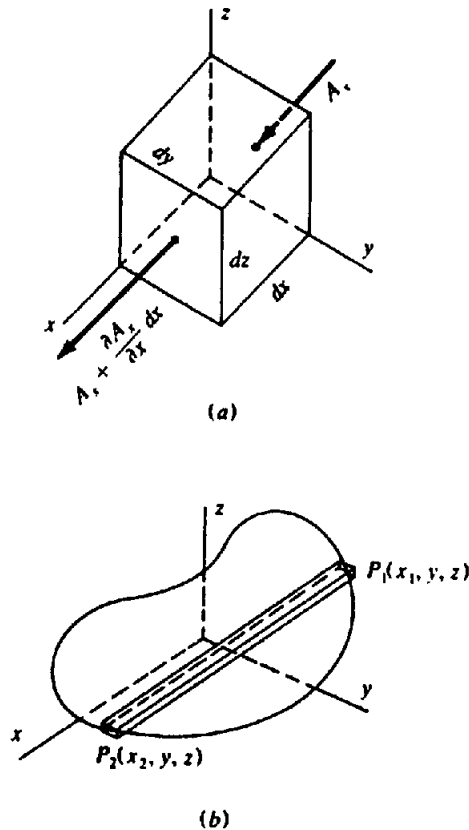


Figure A.2. Vector theorems. (a) Significance of $\text{div } A$. (b) Derivation of Gauss's theorem.

a vector equal to the velocity times density of a fluid. Referring to Figure A.2a, the quantity of fluid entering the element of volume ($dx dy dz$) along the x axis per unit time is ($A_x dy dz$) whereas that leaving the opposite face per unit time is $\{A_x + (\partial A_x / \partial x) dx\} dy dz$. Subtracting, the net outward flow is $(\partial A_x / \partial x) dx dy dz$. Adding terms for the outward flow along the y and z axes and dividing by the volume $dx dy dz$, we find that the net loss per unit volume per unit time is

$$\begin{aligned}\nabla \cdot A &= \text{div } A \\ &= (\partial A_x / \partial x) + (\partial A_y / \partial y) + (\partial A_z / \partial z) \\ &= -(\partial \rho / \partial t)\end{aligned}\quad (\text{A.26})$$

where ρ is the density of the fluid.

The following theorems involve line, surface, and volume integrals. These are discussed in any advanced calculus text, for example, Wiley (1966, pp. 559–67), and so we shall give here only a very terse description.

A *line integral* is written in various ways: $\int_A A \cdot d\mathbf{l}$, $\int_C A \cdot d\mathbf{l}$, $\oint_C A \cdot d\mathbf{l}$, $\oint A \cdot d\mathbf{l}$, C denoting a specific curve and $\oint$ a closed curve; the line integral is the limit of the sum along the curve of products of the element of the curve $d\mathbf{l}$ and the component of A at $d\mathbf{l}$ and parallel to it. A *surface integral*, usually

written $\int_S A \cdot d\mathbf{s}$, $\iint A \cdot d\mathbf{s}$, $\int_S A_n d\mathbf{s}$, or $\iint A_n d\mathbf{s}$, is the limit of the sum over the surface S of products of elements of the surface $d\mathbf{s}$ and the components of A normal to the surface, A_n ; if the surface is closed, A_n is positive when it is in the direction of the outward-drawn normal. A *volume integral*, $\int_V \phi(x, y, z) dv$, $\iiint \phi(x, y, z) dx dy dz$, and so on, is the limit of the sum of products of elements of volume dv and the values of a scalar function $\phi(x, y, z)$ at the centers of dv .

Gauss's theorem, also known as the *divergence theorem*, states that

$$\int_V (\nabla \cdot A) dv = \int_S A \cdot d\mathbf{s} \quad (\text{A.27})$$

where the surface S encloses the volume V . To prove the theorem, we evaluate the first term of the volume integral, $(\partial A_x / \partial x) dx dy dz$, along a strip parallel to the x axis with cross section $dy dz$ and extending from $P_1(x_1, y, z)$ to $P_2(x_2, y, z)$ (Fig. A.2b). Because y and z are fixed along the strip, the result is

$$\begin{aligned}\int_V \left(\frac{\partial A_x}{\partial x} dx \right) dy dz \\ &= \{A_x(x_2, y, z) - A_x(x_1, y, z)\} dy dz \\ &= A_x ds_x|_{P_2} + A_x ds_x|_{P_1}\end{aligned}$$

because $dy dz$ equals ds_x [Eq. (A.12)] at $P_2(x_2, y, z)$ and $-ds_x$ at $P_1(x_1, y, z)$ (note that the outward-drawn normal is along the negative x axis here). Letting y and z vary, the right-hand side becomes the integral $\int_S A_x ds_x$ over the surface S . Adding the integrals along the y and z axes, we obtain Equation (A.27).

Gauss's theorem has an important corollary known as *Green's theorem*. We let A in Equation (A.27) be the vector $w \nabla u$, u and w being scalar functions of position, $u(x, y, z)$, and $w(x, y, z)$. Replacing A in Equation (A.21) with ∇u and ϕ with w , we see that

$$\nabla \cdot (w \nabla u) = w \nabla^2 u + (\nabla w) \cdot (\nabla u)$$

Equation (A.27) now becomes

$$\int_V \{w \nabla^2 u + (\nabla w) \cdot (\nabla u)\} dv = \int_S (w \nabla u) \cdot d\mathbf{s}$$

Interchanging u and w and subtracting the two results gives Green's theorem:

$$\int_V (w \nabla^2 u - u \nabla^2 w) dv = \int_S (w \nabla u - u \nabla w) \cdot d\mathbf{s} \quad (\text{A.28})$$

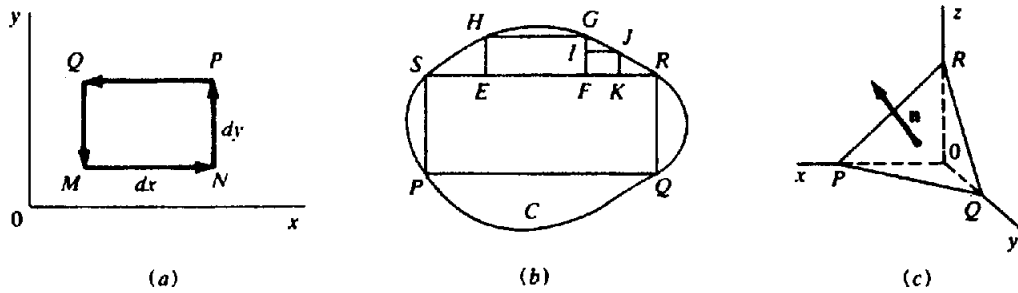


Figure A.3. Derivation of Stokes's theorem. (a) Evaluating a line integral around a rectangle. (b) Evaluating a line integral around an arbitrary closed planar curve C . (c) Resolving a surface integral over a plane triangle into components.

Stokes's theorem relates the line integral of a vector $\mathbf{A}$ along a closed curve C to the surface integral of $\text{curl } \mathbf{A}$ over any open surface that terminates on C . We establish the theorem first for the infinitesimal plane area $dx dy$ in Figure A.3a. We take A_x and A_y as the values of the components at $M(x, y)$. Then the line integral around the rectangle is the sum of the line integrals along the sides MN , NP , PQ , and QM . Thus,

$$\begin{aligned} \int \mathbf{A} \cdot d\mathbf{l} &= A_x dx + \{ A_y + (\partial A_y / \partial x) dx \} dy \\ &\quad - \{ A_x + (\partial A_x / \partial y) dy \} dx - A_y dy \\ &= (\partial A_y / \partial x - \partial A_x / \partial y) dx dy \\ &= (z \text{ component of } \nabla \times \mathbf{A})(ds_z) \quad (\text{A.29a}) \end{aligned}$$

Next we generalize Equation (A.29a) for a plane surface of arbitrary shape (Fig. A.3b). We approximate the area inside C by a large rectangle, then add smaller and smaller rectangles until, in the limit, their sum equals the area inside C . The line integral around the perimeters of the first two rectangles in Figure A.3b is along $PQRFGHESP$ (because EF is traversed twice in opposite directions), and in the limit the path of the line integral will be the curve C and Equation (A.29a) will become

$$\int_C \mathbf{A} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{A})_z ds_z \quad (\text{A.29b})$$

Because Equation (A.29b) holds for a plane figure of any shape, it holds for a plane triangle. Any 3-D surface of arbitrary shape can be approximated as closely as desired by an infinite number of plane triangles. In Figure A.3c, ΔPQR of area ds can be resolved into components [Eq. (A.12)] ΔQRO , ΔRPO , and ΔPQO with areas ds_x , ds_y , and ds_z . The right-hand side of Equation (A.29b) then becomes three surface integrals over these triangles with integrands $(\nabla \times \mathbf{A})_x ds_x$, $(\nabla \times \mathbf{A})_y ds_y$, and $(\nabla \times \mathbf{A})_z ds_z$. When we sum the contributions for all of the triangles making up S , the line integrals along

interior paths cancel in pairs leaving only the line integral along C ; the surface integrals over the triangles can be replaced by integrals over the component triangles perpendicular to the axes, which are then summed at each point and reconstituted into $(\nabla \times \mathbf{A}) \cdot d\mathbf{s}$ for each element of area ds . The result is Stokes's theorem:

$$\int_C \mathbf{A} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s} \quad (\text{A.29c})$$

The positive direction of traversing curve C is the one in which the area is on the left.

If we integrate Equation (A.16) along any curve C joining P_1 and P_2 we get

$$\Delta\phi = \phi_2 - \phi_1 = \int_C \nabla\phi \cdot d\mathbf{l} \quad (\text{A.30})$$

Obviously this result is independent of the path followed. If C is closed, $\phi_2 = \phi_1$ and the integral vanishes. On the other hand, if $\text{curl } \mathbf{A}$ vanishes at all points, the left-hand side of Equation (A.29c) vanishes; therefore $\mathbf{A}$ must be the gradient of a scalar. Thus, if $\text{curl } \mathbf{A} = 0$ everywhere,

$$\mathbf{A} = \nabla\phi \quad (\text{A.31})$$

In this case $\mathbf{A}$ is said to be *irrotational*, or the field of $\mathbf{A}$ is conservative.

A.4. CURVILINEAR COORDINATES

Often we require the functions div , grad , curl , and the Laplacian in cylindrical, spherical, or other coordinates. The first two systems, illustrated in Figure A.4a, b are related to rectangular cartesian coordi-

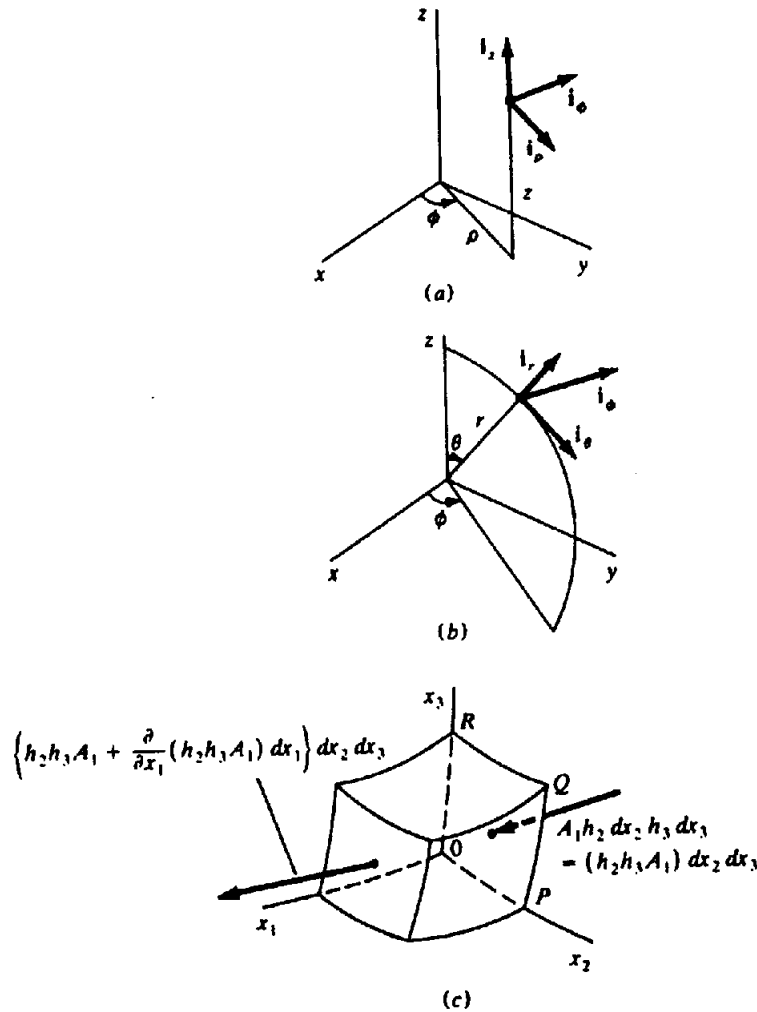


Figure A.4. Curvilinear coordinates. (a) Cylindrical coordinates. (b) Spherical coordinates. (c) $\text{div } A$ in curvilinear coordinates.

nates as follows:

<u>Cylindrical</u>	<u>Spherical</u>	
$x = \rho \cos \phi$	$x = r \sin \theta \cos \phi$	(A.32)
$y = \rho \sin \phi$	$y = r \sin \theta \sin \phi$	
$z = z$	$z = r \cos \theta$	

In cylindrical coordinates we can draw unit vectors i_ρ, i_ϕ, i_z in the directions of increasing ρ, ϕ, z , respectively; for spherical coordinates the unit vectors are i_r, i_θ, i_ϕ . In both cases, the three unit vectors are orthogonal.

If we write x_1, x_2, x_3 for a set of *orthogonal curvilinear coordinates* in general, the coordinate surfaces x_i constant are curved; for example, in spherical coordinates, r constant is a spherical surface, θ constant is a cone, and ϕ constant is a plane. When the coordinate x_i increases by dx_i , a point moves a distance $h_i dx_i$, where h_i is a function of all the x_i 's in general. In cylindrical coordinates, $h_r = 1 = h_z$, and $h_\phi = r$; in spherical coordinates, $h_r = 1$, $h_\theta = r$, and $h_\phi = r \sin \theta$ [because as ϕ changes, r and θ being fixed, a point moves on a circle of radius

$r \sin \theta$, hence a change $d\phi$ moves the point a distance $(r \sin \theta) d\phi$].

The gradient in curvilinear coordinates is obtained by replacing $\partial x, \partial y, \partial z$ in Equation (A.17) by $h_1 \partial x_1, h_2 \partial x_2, h_3 \partial x_3$; the result is

$$\nabla \psi = (\partial \psi / h_1 \partial x_1) i_1 + (\partial \psi / h_2 \partial x_2) i_2 + (\partial \psi / h_3 \partial x_3) i_3 \quad (\text{A.33})$$

To find $\text{div } A$, we use the same concept that we used to get Equation (A.26). Referring to Figure A.4c, the inward flow along the x_1 axis is $A_1(h_2 dx_2 h_3 dx_3) = (h_2 h_3 A_1) dx_2 dx_3$. Subtracting this from the outward flow $[(h_2 h_3 A_1) + \{\partial / \partial x_1 (h_2 h_3 A_1)\} dx_1] dx_2 dx_3$, adding terms for the x_2 and x_3 axes, then dividing by the volume $(h_1 h_2 h_3 dx_1 dx_2 dx_3)$ gives

$$\begin{aligned} \nabla \cdot A &= (1/h_1 h_2 h_3) \\ &\times \{ \partial / \partial x_1 (h_2 h_3 A_1) + \partial / \partial x_2 (h_3 h_1 A_2) \\ &\quad + \partial / \partial x_3 (h_1 h_2 A_3) \} \quad (\text{A.34}) \end{aligned}$$

To get the Laplacian, we find $\nabla^2 = \nabla \cdot \nabla$. Applying Equation (A.34) to Equation (A.33), we get the result

$$\nabla^2 \psi = (1/h_1 h_2 h_3) \left[\frac{\partial}{\partial x_1} \left\{ \left(h_2 h_3 / h_1 \right) \frac{\partial \psi}{\partial x_1} \right\} + \frac{\partial}{\partial x_2} \left\{ \left(h_3 h_1 / h_2 \right) \frac{\partial \psi}{\partial x_2} \right\} + \frac{\partial}{\partial x_3} \left\{ \left(h_1 h_2 / h_3 \right) \frac{\partial \psi}{\partial x_3} \right\} \right] \quad (\text{A.35})$$

We get the expression for curl in the same way that we derived Equation (A.29a). Because the dimensions of face $MNPQ$ in Figure A.4c are infinitesimal, we start with the components A_2 and A_3 at M , then apply Equation (A.29a) for a plane area. This gives

$$\begin{aligned} (\nabla \times \mathbf{A})_1 h_2 dx_2 h_3 dx_3 &= A_2 h_2 dx_2 + \left\{ A_3 h_3 + \frac{\partial}{\partial x_2} (A_3 h_3) dx_2 \right\} dx_3 \\ &\quad - \left\{ A_2 h_2 + \frac{\partial}{\partial x_3} (A_2 h_2) dx_3 \right\} dx_2 \\ &\quad - A_3 h_3 dx_3 \\ &= \left\{ \frac{\partial}{\partial x_2} (A_3 h_3) - \frac{\partial}{\partial x_3} (A_2 h_2) \right\} dx_2 dx_3 \\ (\nabla \times \mathbf{A})_1 &= (1/h_2 h_3) \left\{ \frac{\partial}{\partial x_2} (A_3 h_3) \right. \\ &\quad \left. - \frac{\partial}{\partial x_3} (A_2 h_2) \right\} \end{aligned}$$

When the other two components are found in the same way, the result can be written:

$$\nabla \times \mathbf{A} = (1/h_1 h_2 h_3) \begin{vmatrix} h_1 \mathbf{i}_1 & h_2 \mathbf{i}_2 & h_3 \mathbf{i}_3 \\ \frac{\partial}{\partial x_1} & \frac{\partial}{\partial x_2} & \frac{\partial}{\partial x_3} \\ h_1 A_1 & h_2 A_2 & h_3 A_3 \end{vmatrix} \quad (\text{A.36})$$

Equations (A.33) to (A.36) can be used to obtain grad, div, curl, and the Laplacian in cylindrical and spherical coordinates by inserting the appropriate values of h_i ; for example, the Laplacian in spherical coordinates is, from Equation (A.35),

$$\begin{aligned} \nabla^2 \psi &= \left(\frac{1}{r^2 \sin \theta} \right) \left\{ \frac{\partial}{\partial r} \left(r^2 \sin \theta \frac{\partial \psi}{\partial r} \right) + \frac{\partial}{\partial \theta} \left(\frac{r \sin \theta}{r} \frac{\partial \psi}{\partial \theta} \right) + \frac{\partial}{\partial \phi} \left(\frac{r}{r \sin \theta} \frac{\partial \psi}{\partial \phi} \right) \right\} \\ &= \left(\frac{1}{r^2} \right) \left\{ \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \left(\frac{1}{\sin \theta} \right) \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \left(\frac{1}{\sin^2 \theta} \right) \left(\frac{\partial^2 \psi}{\partial \phi^2} \right) \right\} \quad (\text{A.37}) \end{aligned}$$

Curl $\mathbf{A}$ in spherical coordinates is given by Equation (A.36):

$$\begin{aligned} \nabla \times \mathbf{A} &= (1/r \sin \theta) \left\{ \frac{\partial}{\partial \theta} (\sin \theta A_\phi) - \frac{\partial A_\theta}{\partial \phi} \right\} \mathbf{i}_r \\ &\quad + (1/r \sin \theta) \left\{ \frac{\partial A_r}{\partial \phi} - \sin \theta \frac{\partial (r A_\phi)}{\partial r} \right\} \mathbf{i}_\theta \\ &\quad + (1/r) \left\{ \frac{\partial (r A_\theta)}{\partial r} - \frac{\partial A_r}{\partial \theta} \right\} \mathbf{i}_\phi \quad (\text{A.38}) \end{aligned}$$

A.5. TAYLOR'S SERIES; MACLAURIN'S SERIES

Taylor's series is discussed in most advanced calculus texts, for example, Wiley (1966). One form of the series is

$$\begin{aligned} f(x+h) &= f(x) + hf'(x) + \frac{h^2}{2!} f''(x) \\ &\quad + \cdots + \frac{h^{n-1}}{(n-1)!} f^{n-1}(x) + R(\xi) \end{aligned} \quad (\text{A.39})$$

where $f'(x), f''(x), \dots, f^{n-1}(x)$ are derivatives of order 1, 2, ..., $(n-1)$, respectively, $\xi = kh$, $0 < k < 1$, and $R(\xi)$ is the *remainder after n terms*. The series is valid for all values of h provided the n derivatives exist. Also, $R(\xi) = (h^n/n!)f^n(\xi)$ and approaches zero as n approaches infinity.

Taylor's series enables us to find the change in $f(x)$ when x increases by h in terms of h and the derivatives of $f(x)$. The larger h is, the more terms that we must take to get an accurate value of $f(x+h)$. When h is small, often the first two or three terms are sufficient.

A special case of Taylor's series, called *Maclaurin's series*, is obtained by interchanging x and h in Equation (A.39), then setting h equal to zero. This gives Maclaurin's series:

$$\begin{aligned} f(x) &= f(0) + xf'(0) + \frac{x^2}{2!} f''(0) \\ &\quad + \frac{x^3}{3!} f'''(0) + \cdots \end{aligned} \quad (\text{A.40})$$

where all derivatives are evaluated at $x = 0$. As an example, we calculate the series corresponding to $\sin x$. Because the derivatives of $\sin x$ are either $\pm \sin x$ or $\pm \cos x$ and these equal 0, ± 1 at $x = 0$, the series is

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \quad (\text{A.41a})$$

In the same way, we get for $\cos x$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \quad (\text{A.41b})$$

A.6. BINOMIAL EXPANSION

If we set n equal to 2, 3, 4, and so on, we can show that the general *binomial expansion* of $(a + b)^n$ is

$$\begin{aligned} (a + b)^n &= a^n + na^{n-1}b + \frac{n(n-1)}{2!}a^{n-2}b^2 + \dots \\ &\quad + \frac{n(n-1)\dots(n-r+1)}{r!}a^{n-r}b^r + \dots \end{aligned} \quad (\text{A.42})$$

When n is a positive integer, the series ends with $r = n$, but otherwise the series is infinite. The infinite series is valid provided $|a| > |b|$. (Pipes and Harvill, 1970, p. 843).

We are interested mainly in using Equation (A.42) to find approximate values, so we divide through by a^n and then set $x = b/a$ so that $|x| < 1$. Discarding the factor a^n , the series becomes

$$\begin{aligned} (1 + x)^n &= 1 + nx + \frac{n(n-1)}{2!}x^2 \\ &\quad + \frac{n(n-1)(n-2)}{3!}x^3 + \dots \end{aligned} \quad (\text{A.43})$$

Note that the series converges (has a finite sum) for all values of n (Pipes and Harvill, 1970, p. 843).

The series for $n = -1, -2$ are especially useful:

$$(1 \mp x)^{-1} = 1 \pm x + x^2 \pm x^3 + \dots \quad (\text{A.44a})$$

$$(1 \mp x)^{-2} = 1 \pm 2x + 3x^2 \pm 4x^3 + \dots \quad (\text{A.44b})$$

A.7. COMPLEX NUMBERS

Because we cannot calculate the square root of a negative number, a quantity such as $\sqrt{-7}$ is said to be *imaginary*. Writing $j = \sqrt{-1}$ (i is also used), imaginary numbers can be written jy (or yi), y being real; thus, $\sqrt{-7} = j\sqrt{7}$. The sum of a real number and an imaginary number is a *complex number*, for example, $3 - 8j$. The *conjugate complex* of $z = (x + jy)$ is $\bar{z} = (x - jy)$.

A complex quantity $f(z)$ can be represented geometrically in the *complex plane*; this is merely the xy plane with the real part, $\text{Re}\{f(z)\}$, of a complex quantity plotted on the x axis, the imaginary part, $\text{Im}\{f(z)\}$, on the y axis (Fig. A.5). Plotted thus,

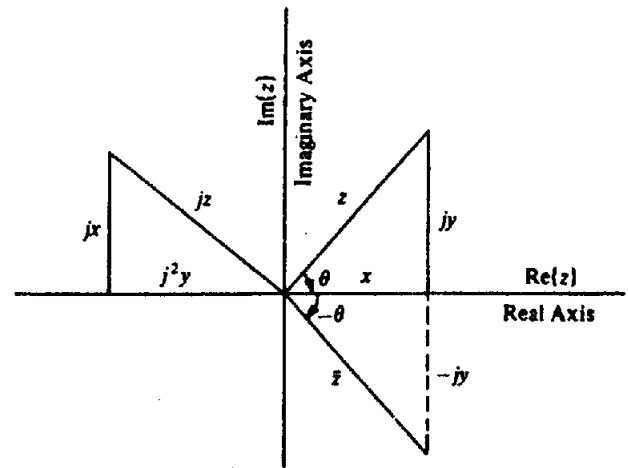


Figure A.5. Plotting complex quantities in the complex plane.

complex numbers are similar to 2-D vectors, real and imaginary parts corresponding to components. Complex numbers are added (or subtracted) by adding (subtracting) separately the real and imaginary parts; thus,

$$(4 - 7j) + (6 + 4j) - (18 - 12j) = (-8 + 9j)$$

Multiplication of complex numbers follows the usual rules of algebra, keeping in mind that $j^2 = -1$, $j^3 = -j$, $j^4 = +1$, and so on; thus,

$$\begin{aligned} (8 - 9j)(-4 - 6j) &= -32 + 36j - 48j + 54j^2 \\ &= -86 - 12j \end{aligned}$$

Multiplication by j rotates z and its real and imaginary components through 90° counterclockwise (Fig. A.5). Multiplication of z by $\bar{z}$ gives $(x^2 + y^2)$; the square root of this quantity is the magnitude of z , $|z|$; thus $|z|$ is the length of the line representing z in Figure A.5.

Division is done by multiplying the numerator and denominator by the conjugate complex of the denominator (called *rationalization of the denominator*), then dividing; for example,

$$\begin{aligned} \frac{(7 - 2j)}{(-9 + 5j)} &= \frac{(7 - 2j)(-9 - 5j)}{(-9 + 5j)(-9 - 5j)} \\ &= \frac{-(73 + 17j)}{(9^2 + 5^2)} = \frac{-(73 + 17j)}{106} \end{aligned}$$

A complex quantity is zero only when its real and imaginary parts are zero. It follows that two complex quantities are equal only when their real parts are equal and their imaginary parts are also equal.

The base of the system of natural logarithms, e , is defined by the infinite series, valid for all values of x ,

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \quad (\text{A.45a})$$

Setting $x = 0$, we get a series from which the value of e can be calculated. Replacing x with $\pm jx$, we get the following series:

$$e^{+jx} = 1 + jx - (x^2/2!) - j(x^3/3!) + \dots \quad (\text{A.45b})$$

$$e^{-jx} = 1 - jx - (x^2/2!) + j(x^3/3!) + \dots \quad (\text{A.45c})$$

Comparing these series with Equations (A.41), we find that

$$\left. \begin{aligned} \cos x &= \frac{1}{2}(e^{jx} + e^{-jx}) \\ \sin x &= (1/2j)(e^{jx} - e^{-jx}) \end{aligned} \right\} \quad (\text{A.46a})$$

$$\left. \begin{aligned} (\cos x + j \sin x) &= e^{jx} \\ (\cos x - j \sin x) &= e^{-jx} \end{aligned} \right\} \quad (\text{A.46b})$$

Replacing x by nx , we get *De Moivre's theorem*:

$$(\cos nx \pm j \sin nx) = e^{\pm jnx} = (\cos x \pm j \sin x)^n \quad (\text{A.46c})$$

When $z = x + jy$, we define θ by the relation (Fig. A.5)

$$\theta = \tan^{-1}(y/x) = \tan^{-1}[\mathcal{I}\{z\}/\mathcal{R}\{z\}] \quad (\text{A.47a})$$

where θ is the phase of z . Also,

$$z = |z|(\cos \theta + j \sin \theta) = |z|e^{j\theta} \quad (\text{A.47b})$$

using Equation (A.46b); this is known as the *polar form* of z . Changing the sign of $j \sin \theta$ and using Equation (A.46b) again, we obtain

$$\bar{z} = |z|e^{-j\theta} \quad (\text{A.47c})$$

The polar form of z is convenient for multiplication, division, and finding powers and roots; for example, if $z_1 = |z_1|e^{j\theta_1}$ and $z_2 = |z_2|e^{j\theta_2}$, then

$$z_1 z_2 = |z_1||z_2|e^{j(\theta_1 + \theta_2)}$$

$$z_1/z_2 = (|z_1|/|z_2|)e^{j(\theta_1 - \theta_2)}$$

$$z_1^5 = |z_1|^5 e^{j5\theta_1}$$

$$z_1^{1/4} = |z_1|^{1/4} e^{j(\theta_1 + 2\pi r)/4} \quad r = 0, 1, 2, 3$$

(In finding the fourth root of z , insertion of $2\pi r$ gives four distinct roots; these repeat when $r > 3$.)

A.8. METHOD OF LEAST SQUARES

Assume that we measure a quantity y for various values of x and end up with a set of values (y_i, x_i) ; often we wish to express y as a function of x in the form

$$y = a_0 + a_1x + a_2x^2 + \dots + a_mx^m \quad (\text{A.48})$$

In principle, we could solve for the $(m+1)$ unknowns a_i if we had $(m+1)$ sets of values (y_i, x_i) , but usually we have n sets, $n > (m+1)$, so the constants a_i are "overdetermined." In this case we look for the "best-fit" solution; the usual criterion for best fit is that the sums of the squares of the "errors" be a minimum, an error being the difference between a measured value of y_i and the value calculated from Equation (A.48) for the corresponding value x_i and the best-fit values of a_i . Squares of the errors are used to avoid cancellation of positive and negative errors.

The sum of the errors squared is

$$\begin{aligned} E &= \sum_i e_i^2 \\ &= \sum_i \{ y_i - (a_0 + a_1x_i + a_2x_i^2 + \dots + a_mx_i^m) \}^2 \end{aligned} \quad (\text{A.49})$$

To get the minimum of E , we vary the $(m+1)$ a_i 's; hence for each a_i ,

$$\partial E / \partial a_r = 0 = \sum_i \{ y_i - (a_0 + a_1x_i + a_2x_i^2 + \dots + a_mx_i^m) \} (-2x_i^r)$$

so that

$$\begin{aligned} a_0 \sum_i x_i^r + a_1 \sum_i x_i^{r+1} + a_2 \sum_i x_i^{r+2} \\ + \dots + a_m \sum_i x_i^{r+m} = \sum_i y_i x_i^r \end{aligned} \quad (\text{A.50})$$

where $r = 0, 1, 2, \dots, m$ (note that for $r = 0$, $\sum_i x_i^0 = n$ and $\sum_i y_i x_i^0 = \sum_i y_i$). This set of $(m+1)$ equations can be solved for the $(m+1)$ a_i 's.

For a more detailed discussion, including the matrix solution of Equation (A.50), see Sheriff and Geldart (1983, §10.1.5).

A.9. FOURIER SERIES AND TRANSFORMS

A.9.1. Fourier Series

A periodic function $f(t)$, which repeats itself exactly after each interval T , can (with few exceptions) be represented by a *Fourier series* of the form

$$\begin{aligned} g(t) = & a_0 + a_1 \cos \omega_0 t + a_2 \cos 2\omega_0 t \\ & + \cdots + a_n \cos n\omega_0 t + \cdots + b_1 \sin \omega_0 t \\ & + b_2 \sin 2\omega_0 t + \cdots + b_n \sin n\omega_0 t + \cdots \end{aligned} \quad (\text{A.51a})$$

where $\omega_0 = 2\pi\nu_0 = 2\pi/T$ is the *fundamental angular frequency* and t is the independent variable (often time or distance). The frequency $n\omega_0$ is the n th *harmonic* of the fundamental ω_0 , and so expressing a function as a Fourier series is called *harmonic analysis* (the same name is applied to transforming a function into its Fourier transform; see §A.9.2).

The integral over the interval T of the product of any two cosines, two sines, or sine and cosine in Equation (A.51a) is zero except for the integrals of $\cos^2 n\omega_0 t$ and $\sin^2 n\omega_0 t$, and these both equal $T/2$. Thus, to find the values of a_n and b_n we multiply both sides of Equation (A.51a) by $\cos n\omega_0 t$ to get a_n , then by $\sin n\omega_0 t$ to get b_n . Integration over the interval T results in

$$\left. \begin{aligned} a_n &= (2/T) \int_{-T/2}^{T/2} g(t) \cos n\omega_0 t \, dt \\ b_n &= (2/T) \int_{-T/2}^{T/2} g(t) \sin n\omega_0 t \, dt \end{aligned} \right\} \quad (\text{A.51b})$$

In practice Fourier series are used to analyze a set of observations, such as gravity readings along a profile; the integrals become sums, the length of the profile is T (note that this means that we are assuming that the readings will be repeated exactly if the profile is extended), and the relative values of a_n , b_n [more accurately, $(a_n^2 + b_n^2)^{1/2}$] are interpreted as *magnitudes* of the n th harmonic.

Fourier series can be developed for more than one dimension. The 2-D form is

$$\begin{aligned} g(x, y) = & \sum_0^\infty \sum_0^\infty (a_{mn} \cos m\omega_x x \cos n\omega_y y \\ & + b_{mn} \cos m\omega_x x \sin n\omega_y y \\ & + c_{mn} \sin m\omega_x x \cos n\omega_y y \\ & + d_{mn} \sin m\omega_x x \sin n\omega_y y) \end{aligned} \quad (\text{A.52a})$$

where the coefficients are given by equations such as

$$\begin{aligned} b_{mn} = & (4/T_x T_y) \\ & \times \int_{-T/2}^{T/2} \int_{-T/2}^{T/2} g(x, y) \cos m\omega_x x \sin n\omega_y y \, dx \, dy \end{aligned} \quad (\text{A.52b})$$

Equation (A.52a) defines a *Fourier surface* that can be fitted to a two-dimensional set of data, for example, gravity readings over an area, by finding the four sets of coefficients using double sums equivalent to the integral in Equation (A.52b).

A.9.2. Fourier Integral; Fourier Transforms

Equation (A.46a) can be used to write Equations (A.51) as

$$\begin{aligned} g(t) &= \sum_{-\infty}^{+\infty} \alpha_n e^{jn\omega_0 t} \\ \alpha_n &= (1/T) \int_{-T/2}^{T/2} g(t) e^{-jn\omega_0 t} \, dt \\ n &= 0, \pm 1, \pm 2, \dots, \pm \infty \end{aligned} \quad (\text{A.53})$$

Combining the two relations gives

$$g(t) = \sum_{-\infty}^{+\infty} (1/T) e^{jn\omega_0 t} \int_{-T/2}^{T/2} g(t) e^{-jn\omega_0 t} \, dt$$

If we let T approach infinity, $g(t)$ repeats itself at longer and longer intervals; in the limit when $T = \infty$, $g(t)$ does not repeat, hence is aperiodic. The preceding expression then becomes the *Fourier integral*:

$$g(t) = (1/2\pi) \int_{-\infty}^{\infty} e^{j\omega t} \left\{ \int_{-\infty}^{\infty} g(t) e^{-j\omega t} \, dt \right\} d\omega \quad (\text{A.54})$$

(see Wiley, 1966, §6.7, for more details). If we evaluate the integral within the braces, we get a function of ω ; then the integral with respect to ω gives us $g(t)$ again (except for the factor $1/2\pi$).

The Fourier integral is easier to apply if we break Equation (A.54) down into two operations:

$$\begin{aligned} G(\omega) &= \int_{-\infty}^{\infty} g(t) e^{-j\omega t} \, dt \\ &= \text{Fourier transform of } g(t) \end{aligned} \quad (\text{A.55a})$$

$$\begin{aligned} g(t) &= (1/2\pi) \int_{-\infty}^{\infty} G(\omega) e^{j\omega t} \, d\omega \\ &= \text{inverse Fourier transform of } G(\omega) \end{aligned} \quad (\text{A.55b})$$

When t is time, application of Equation (A.55a)

transforms $g(t)$ from the *time domain* into the *frequency domain* whereas Equation (A.55b) does the reverse. (The terms *time domain* and *frequency domain* are often used even when t and ω are not time and frequency.) The distinction between *transform* and *inverse transform* is arbitrary and is due to our greater familiarity with the time domain.

The Fourier transform has several advantages in data processing. Equations (A.55) show that $g(t)$ can be transformed into the frequency domain, then back to the time domain unchanged, that is, theoretically there is no error involved [in practice the integrals must be calculated numerically and so errors (loss of information) exist, but these can be made as small as we wish]. A major advantage is that we can do part of the processing in one domain and part in the other, taking advantage of the fact that some processes can be executed more economically in one domain than in the other.

The functions $g(t)$ and $G(\omega)$ [$G(\nu)$ is sometimes used, ω being replaced by $2\pi\nu$ in the exponentials in Eqs. (A.55), $d\omega/2\pi$ becoming $d\nu$] are referred to as a *transform pair*. The relation between them can be expressed as $g(t) \leftrightarrow G(\omega)$. In general, $G(\omega)$ is complex and can be written in the form [Eq. (A.47b)]

$$G(\omega) = A(\omega) e^{j\phi(\omega)} \quad (\text{A.56})$$

$A(\omega)$ is the *frequency spectrum* and $\phi(\omega)$ is the *phase spectrum* of $g(t)$.

Like the Fourier series, Fourier transforms can be used with functions of several dimensions. In two dimensions, we have

$$G(\kappa_x, \kappa_y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) e^{-j(\kappa_x x + \kappa_y y)} dx dy, \quad (\text{A.57a})$$

$$g(x, y) = (1/2\pi)^2 \times \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} G(\kappa_x, \kappa_y) e^{j(\kappa_x x + \kappa_y y)} d\kappa_x d\kappa_y \quad (\text{A.57b})$$

where $\kappa_x = 2\pi/\lambda_x$ and $\kappa_y = 2\pi/\lambda_y$ (compare with $\omega = 2\pi/T$).

Calculation of Fourier transforms of continuous functions is often quite difficult; however, in practice we use digital functions mainly (§A.9.3) and their transforms are easy to calculate. Calculation of both types is greatly facilitated by using several theorems regarding Fourier transforms; we list the most important of these in Equations (A.58) to (A.62) and give brief explanations of their applications. Other theorems plus proofs can be found in Sheriff and Geldart (1983, pp. 164–6).

Shift theorems:

$$g(t - k) \leftrightarrow e^{-j\omega k} G(\omega) \quad (\text{A.58a})$$

$$e^{-jkt} g(t) \leftrightarrow G(\omega + k) \quad (\text{A.58b})$$

Scaling theorem:

$$g(kt) \leftrightarrow (1/|k|) G(\omega/k) \quad (\text{A.59})$$

Symmetry theorem:

$$G(t) \leftrightarrow 2\pi g(-\omega) \quad (\text{A.60})$$

Derivative theorems:

$$\frac{d^n g(t)}{dt^n} \leftrightarrow (j\omega)^n G(\omega) \quad (\text{A.61a})$$

$$(-jt)^n g(t) \leftrightarrow \frac{d^n G(\omega)}{d\omega^n} \quad (\text{A.61b})$$

Integral theorem:

$$\int_{-\infty}^t g(t) dt \leftrightarrow (1/j\omega) G(\omega) \quad (\text{A.62})$$

When we add a constant to the independent variable, functions of the variable are shifted along the horizontal axis; for example, the curve $y = (x - 3)^2$ is the curve $y = x^2$ shifted three units to the right; $y = (x + 3)^2$ is $y = x^2$ moved to the left three units. Equation (A.58a) states that shifting $g(t)$ k units to the right multiplies the transform by $\exp(-j\omega k)$; shifting $G(\omega)$ k units to the left multiplies $g(t)$ by $\exp(-jkt)$. Also, if we know $G(\omega)$, we can write down the transform of $e^{-jkt} g(t)$ by inspection.

Equation (A.59) permits us to replace t with, for example, $-7t/8$, without having to recalculate the transform. Equation (A.60) is useful in obtaining new transforms; if we know a transform pair, $g(t)$ and $G(\omega)$, we get a new transform pair by replacing ω in $G(\omega)$ with t and t in $g(t)$ with $-\omega$.

The derivative theorems are probably the most important theorems. Equation (A.61a) enables us to replace transforms of derivatives of $g(t)$ with $(j\omega)^n$ times the transform of $g(t)$; this is very useful in finding new transforms and in replacing differential equations with algebraic equations (§A.13). Equation (A.61b) is useful in getting transforms of $t^n g(t)$. Equation (A.62), although used infrequently, is nevertheless important; note that for causal functions (§A.9.3), the lower limit of the integral is zero.

A.9.3. Digital Functions; z Transforms

A continuous function $g(t)$ has a definite value for all values of t (disregarding discontinuities, infinite and multiple values), whereas a *digital function*, which we write as g_i , has values only at discrete values of t ; for example, a seismic trace (§4.1.3) may be sampled at intervals $\Delta = 0.002$ s so that g_i is known only at the discrete times 0.000, 0.002, 0.004, ... s, or we might measure gravity values along a line at intervals of 100 m so that g_x is known only at points 100 m apart. (Whenever convenient, we take $\Delta = 1$, that is, we take it as the unit so that $t = n$ or $x = n$ instead of $n\Delta$.)

The *unit impulse* or *Dirac delta*, $\delta(t)$ or δ_i , is by definition zero for all values of t except $t = 0$ where $\delta(t)$ (or δ_i) equals +1 (for a more rigorous definition, see Papoulis, 1962, Appendix I). The product $g(t)\delta(t) = g(0)$, the value of $g(t)$ at $t = 0$. The shifted impulse $\delta(t - n)$ or δ_{i-n} is zero except at $t = n$. A *comb* is an infinite series of unit impulses spaced at intervals Δ ; thus,

$$\text{comb}(t) = \sum_{-\infty}^{\infty} \delta(t - n) \\ n = 0, \pm 1, \pm 2, \pm 3, \dots, \pm \infty$$

Multiplying $g(t)$ by $\text{comb}(t)$ gives g_i :

$$g_i = \sum_{-\infty}^{\infty} g(t) \delta(t - n) \\ = \dots g_{-3}, g_{-2}, g_{-1}, g_0, g_1, g_2, g_3, \dots$$

Normally the digital functions that we encounter are *causal*, that is, they are zero for negative values of t ; in this case the first term is g_0 .

If we substitute $\delta(t)$ for $g(t)$ in Equations (A.55a) and (A.58a), we find that

$$\left. \begin{aligned} \delta(t) &\leftrightarrow e^{-j\omega t} \Big|_{t=0} = +1 \\ \delta(t - n) &\leftrightarrow e^{-jn\omega\Delta} \end{aligned} \right\} \quad (\text{A.63})$$

Accordingly, the transform of g_i is

$$g_i \leftrightarrow g_0 + g_1 e^{-j\omega\Delta} + g_2 e^{-j2\omega\Delta} + \dots + g_n e^{-jn\omega\Delta} \\ \leftrightarrow g_0 + g_1 z + g_2 z^2 + \dots + g_n z^n \quad (\text{A.64})$$

where $z = e^{-j\omega\Delta}$. The right-hand side of Equation (A.64) is $G(z)$, the z transform of g_i . Obviously, calculation of z transforms is trivial, for example, for a causal function $g_i = 0.56, -1.24, -2.45, 0.67, 3.78$, the z transform is $0.56 - 1.24z - 2.45z^2 + 0.67z^3 + 3.78z^4$.

In data processing, n in Equation (A.64) may be several hundred or thousand; at the same time we

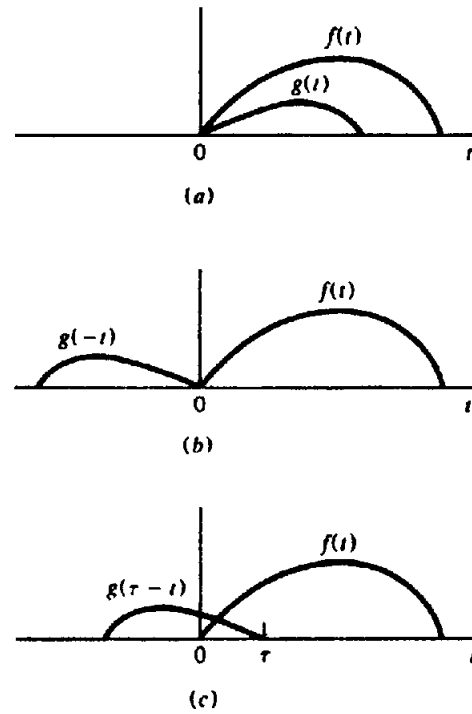


Figure A.6. Illustrating $f(t) * g(t)$. (a) Two functions, $f(t)$ and $g(t)$. (b) $g(t)$ reflected in the vertical axis. (c) $g(t)$ reflected and displaced τ units to the right.

often need values of $G(z)$ for a large range of z values (that is, ω values); if we adopt the straightforward approach, the calculations are extremely laborious. However, the method known as the *fast Fourier transform* (FFT) reduces the labor by a factor of 10^3 or more when $n > 10^3$ [Sheriff and Geldart, 1983, §10.6.4].

A.10. CONVOLUTION

The *convolution* of two functions, $f(t)$ and $g(t)$, often written $f(t) * g(t)$, is defined by the integral

$$f(\tau) * g(\tau) = \int_{-\infty}^{\infty} f(t) g(\tau - t) dt \quad (\text{A.65a})$$

The curve $g(-t)$ is $g(t)$ reflected in the vertical axis (Fig. A.6a, b) and $g(\tau - t)$ is $g(-t)$ moved τ units to the right (Fig. A.6c). Thus, geometrically, Equation (A.65a) means that we reflect $g(t)$, move it τ units to the right, and then sum the products of the corresponding ordinates. The result depends on the displacement τ , hence the argument τ on the left side of Equation (A.65a). For causal functions of length m and n , it is obvious that $0 \leq \tau \leq (m + n)$. Also, we get the same result if we reflect and displace $f(t)$ instead of $g(t)$, that is,

$$f(t) * g(t) = g(t) * f(t) \quad (\text{A.66})$$

The *convolution theorem* states that

$$f(\tau) * g(\tau) \leftrightarrow F(\omega) G(\omega) \quad (\text{A.67a})$$

To prove the theorem, we write

$$\begin{aligned} f(\tau) * g(\tau) &\leftrightarrow \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} f(t) g(\tau - t) dt \right\} e^{-j\omega\tau} d\tau \end{aligned}$$

using Equation (A.55a) with τ in place of t [because $f(\tau) * g(\tau)$ is a function of τ]. Interchanging the order of integration and writing $s = \tau - t$ and $ds = d\tau$ (because t is fixed in this integration), we obtain

$$\begin{aligned} f(\tau) * g(\tau) &\leftrightarrow \int_{-\infty}^{\infty} f(t) \left\{ \int_{-\infty}^{\infty} g(s) e^{-j\omega(s+t)} ds \right\} dt \\ &\leftrightarrow \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt \int_{-\infty}^{\infty} g(s) e^{-j\omega s} ds \\ &\leftrightarrow F(\omega) G(\omega) \end{aligned}$$

For digital functions, Equations (A.65a), (A.66), and (A.67a) are

$$f_{\tau} * g_{\tau} = \sum_k f_k g_{\tau-k} = \sum_k g_k f_{\tau-k} \quad (\text{A.65b})$$

$$f_{\tau} * g_{\tau} \leftrightarrow F(z) G(z) \quad (\text{A.67b})$$

where the sums in Equation (A.65b) are over the appropriate values of k [$0 \leq k \leq (m+n)$].

In two dimensions the convolution of $f(x, y)$ and $g(x, y)$ becomes (Sheriff and Geldart, 1983, §10.3.9)

$$\begin{aligned} f(\tau, \sigma) * g(\tau, \sigma) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) g(\tau - x, \sigma - y) dx dy \\ &\quad (\text{A.68}) \end{aligned}$$

and the convolution theorem states that

$$f(\tau, \sigma) * g(\tau, \sigma) \leftrightarrow F(\kappa_x, \kappa_y) G(\kappa_x, \kappa_y) \quad (\text{A.69})$$

A.11. CORRELATION

A.11.1. Cross-Correlation

The *cross-correlation* of $f(t)$ and $g(t)$, $\phi_{fg}(\tau)$, is defined as

$$\phi_{fg}(\tau) = \int_{-\infty}^{\infty} f(t) g(t + \tau) dt \quad (\text{A.70a})$$

If we find the convolution of $f(t)$ and $g(-t)$, we change $g(\tau - t)$ in Equation (A.65a) to $g(\tau + t)$;

therefore

$$\phi_{fg}(\tau) = \text{convolution of } f(t) \text{ with } g(-t) \quad (\text{A.71a})$$

$$\phi_{gf}(\tau) = \text{convolution of } f(-t) \text{ with } g(t) \quad (\text{A.71b})$$

The *cross-correlation theorem* states that

$$\begin{aligned} \phi_{fg}(\tau) &\leftrightarrow \overline{F(\omega)} G(\omega) \\ \phi_{gf}(\tau) &\leftrightarrow F(\omega) \overline{G(\omega)} \end{aligned} \quad (\text{A.72a})$$

The proof follows directly from the proof of the convolution theorem on changing the sign of t in the argument of $g(\tau - t)$ in the first step of the proof.

The digital forms of Equations (A.70a) and (A.72a) are

$$\phi_{fg} = \sum_k f_k g_{\tau+k} \quad \phi_{gf} = \sum_k f_{\tau+k} g_k \quad (\text{A.70b})$$

where $-m < k < n$ or $-n < k < m$.

$$\begin{aligned} \phi_{fg} &\leftrightarrow F(\bar{z}) G(z) \\ \phi_{gf} &\leftrightarrow F(z) G(\bar{z}) \end{aligned} \quad (\text{A.72b})$$

where $\bar{z} = e^{+j\omega\Delta}$ and $F(\bar{z}) = \overline{F(z)}$.

A.11.2. Autocorrelation

If $f(t)$ is the same as $g(t)$, the cross-correlation becomes the *autocorrelation*. Replacing $f(t)$ with $g(t)$ in the preceding section, we get the following results.

$$\phi_{gg}(\tau) = \int_{-\infty}^{\infty} g(t) g(t + \tau) dt \leftrightarrow |G(\omega)|^2 \quad (\text{A.73})$$

$$\phi_{gg} = \sum_k g_k g_{\tau+k} \quad \phi_{gg} \leftrightarrow |G(z)|^2 \quad (\text{A.74})$$

A.12. LAPLACE TRANSFORMS

A.12.1. Basic Theory

Laplace transforms are closely related to Fourier transforms (Sheriff and Geldart, 1983, pp. 172-4). The commonly used *Laplace transform* is *one-sided*, that is, the transform integral has the lower limit $t = 0$, so that all functions are in effect causal. The transform pair $g(t) \leftrightarrow G(s)$ is defined by the relations

$$G(s) = \int_0^{\infty} g(t) e^{-st} dt \quad (\text{A.75a})$$

$$g(t) = (1/2\pi j) \int_{\sigma-j\omega}^{\sigma+j\omega} G(s) e^{st} ds \quad (\text{A.75b})$$

where $s = \sigma + j\omega$, σ is positive and large enough that $\lim_{t \rightarrow \infty} g(t)e^{-\sigma t} = 0$ (in practice almost all functions satisfy this requirement). Generally $G(s)$ is more easily calculated than $G(\omega)$, but the inverse transformation of Equation (A.75b) involves integration in the complex plane, which is usually difficult. Fortunately, many extensive tables of Laplace transforms are readily available (for example, Pipes and Harvill, 1970, Appendix A), so we get transform pairs from tables, just as in the case of integrals.

Most theorems for Fourier transforms have counterparts for Laplace transforms. The most useful are listed in the following equations (proofs can be found in Sheriff and Geldart, 1983, pp. 173-4):

Shift theorems:

$$g(t-a) \leftrightarrow e^{-as}G(s) \quad (\text{A.76a})$$

$$e^{-at}g(t) \leftrightarrow G(s+a) \quad (\text{A.76b})$$

Scaling theorem:

$$g(at) \leftrightarrow (1/a)G(s/a) \quad (\text{A.77})$$

Derivative theorems:

$$\begin{aligned} d^n g(t)/dt^n &\leftrightarrow s^n G(s) - s^{n-1}g(0+) \\ &\quad - s^{n-2}g'(0+) \\ &\quad - s^{n-3}g''(0+) - \dots - g^{(n-1)}(0+) \end{aligned} \quad (\text{A.78a})$$

$$(-t)^n g(t) \leftrightarrow d^n G(s)/ds^n \quad (\text{A.78b})$$

The notation $g(0+)$ and $g'(0+)$ denotes $g(t)$ and $d'g(t)/dt'$ evaluated at $t=0$ where t approaches zero from the positive side (this allows for discontinuities at $t=0$).

Fourier transforms are convenient for studying the frequency and phase characteristics of a function whereas the Laplace transform is more useful in studying the analytical properties of the transform (as in filter design). Both are useful in solving differential equations with constant coefficients, the choice depending mainly on the type of boundary conditions.

A.12.2. Calculation of Laplace Transforms

Many transforms are easily found using Equation (A.75a); for example,

$$\cos at \leftrightarrow s/(s^2 + a^2)$$

$$\sin at \leftrightarrow a/(s^2 + a^2)$$

$$t^n \leftrightarrow n!/s^{n+1}$$

$$te^{at} \leftrightarrow 1/(s-a)^2$$

$$t \sin at \leftrightarrow 2as/(s^2 + a^2)^2$$

Many transforms can be found by using the theorems, for example, the transform of $\sin at$ can be found by differentiating $\cos at$; te^{at} can be found by applying Equation (A.76b) to the transform of t^n with $n=1$; $t \sin at$ can be found using (A.78b).

Two useful "special" functions are the unit impulse, $\delta(t)$, introduced in Section A.9.3, and the unit step function, $u(t)$, defined as

$$u(t) = \begin{cases} 0 & t \leq 0 \\ +1 & t \geq 0 \end{cases} \quad (\text{A.79})$$

The step function is discontinuous at $t=0$; it is often used to "wipe out" a function for $t < 0$; thus,

$$\begin{aligned} \cos at u(t) &= 0 & t \leq 0 \\ &= \cos at & t \geq 0 \end{aligned}$$

The transforms of $\delta(t)$ and $u(t)$ are easily found from Equation (A.75a):

$$\delta(t) \leftrightarrow +1 \quad u(t) \leftrightarrow 1/s \quad (\text{A.80})$$

A.12.3. Transforms of the Error Function and its Derivatives

The error function, $\text{erf}(x)$, is defined as

$$\text{erf}(x) = (1/\pi^{1/2}) \int_0^x e^{-u^2} du \quad (\text{A.81})$$

We also use $\text{erfc}(x) = 1 - \text{erf}(x)$ which is the complementary error function. From the definition, $\text{erf}(0) = 0$, $\text{erf}(+\infty) = 1$. The error function is tabulated in many handbooks (for example, Gradshteyn and Ryzhik, 1965).

Derivatives of $\text{erf}(x)$ can be obtained from Equation (A.81). Thus,

$$\text{erf}'(x) = d/dx \{ \text{erf}(x) \} = (2/\pi^{1/2}) e^{-x^2} \quad (\text{A.82a})$$

on using Leibnitz' rule (Wiley, 1966, p. 274; Kaplan, 1952, p. 218). If x is replaced by $(\alpha/2t^{1/2})$ [Eq. (7.31)] and we write $\text{erf}'(\alpha/2t^{1/2})$ for the derivative of $\text{erf}(\alpha/2t^{1/2})$ with respect to t , then on setting $y = (\alpha/2t^{1/2})$, we find

$$\begin{aligned} \text{erf}'(\alpha/2t^{1/2}) &= d/dy \{ \text{erf}(y) \} dy/dt \\ &= (2/\pi^{1/2}) e^{-y^2} (-\alpha/4t^{3/2}) \\ &= (-\alpha/2\pi^{1/2}t^{3/2}) e^{-\alpha^2/4t} \quad (\text{A.82b}) \end{aligned}$$

We can expand $\operatorname{erf}(x)$ in a power series, one form being (Pipes and Harvill, 1970, p. 493)

$$\begin{aligned}\operatorname{erf}(x) &= (2/\pi^{1/2}) \{ x - x^3/3 \times 1! \\ &\quad + x^5/5 \times 2! - \dots \\ &\quad + (-1)^n x^{2n+1}/(2n+1) \times n! + \dots \} \quad (\text{A.83})\end{aligned}$$

The Laplace transform of $\operatorname{erfc}(\alpha/2t^{1/2})$ is listed in most tables (Pipes and Harvill, 1970, p. 778):

$$\operatorname{erfc}(\alpha/2t^{1/2}) u(t) \leftrightarrow e^{-\alpha s^{1/2}}/s \quad (\text{A.84a})$$

hence

$$\operatorname{erf}(\alpha/2t^{1/2}) u(t) \leftrightarrow (1 - e^{-\alpha s^{1/2}})/s \quad (\text{A.84b})$$

on using Equation (A.80) [$u(t)$ is usually omitted in tables].

Although α is a constant in Equation (A.84), we can fix t and vary α ; then

$$\partial/\partial\alpha = \partial/\partial y (\partial y/\partial\alpha) = (1/2t^{1/2}) \partial/\partial y$$

Writing Equation (A.84b) in the form of Equation (A.75) and differentiating with respect to α under the integral sign, we obtain

$$\begin{aligned}\int_0^t (1/2t^{1/2}) \operatorname{erf}'(\alpha/2t^{1/2}) e^{-st} dt \\ = d/d\alpha \{ (1 - e^{-\alpha s^{1/2}})/s \} = e^{-\alpha s^{1/2}}/s^{1/2}\end{aligned}$$

Comparing with Equation (A.75) we see that

$$(1/2t^{1/2}) \operatorname{erf}'(\alpha/2t^{1/2}) \leftrightarrow e^{-\alpha s^{1/2}}/s^{1/2} \quad (\text{A.85a})$$

Using Equation (A.82a), Equation (A.85a) becomes

$$(1/\pi t)^{1/2} e^{-\alpha^2/4t} \leftrightarrow e^{-\alpha s^{1/2}}/s^{1/2}$$

Differentiating again with respect to α , we get

$$(\alpha/2\pi^{1/2}t^{3/2}) e^{-\alpha^2/4t} \leftrightarrow e^{-\alpha s^{1/2}}$$

or, on using Equation (A.82a),

$$(\alpha/4t^{3/2}) \operatorname{erf}'(\alpha/2t^{1/2}) \leftrightarrow e^{-\alpha s^{1/2}} \quad (\text{A.85b})$$

Equations (A.85) and (A.84) comprise a series of transform pairs with descending powers of s . The following two pairs, found in Abramowitz and

Stegun (1972) and Erdélyi (1954) respectively, extend this series:

$$e^{-\alpha s^{1/2}}/s^{3/2} \leftrightarrow 2(t/\pi)^{1/2} e^{-\alpha^2/4t} - \alpha \operatorname{erfc}(\alpha/2t^{1/2}) \quad (\text{A.85c})$$

$$\begin{aligned}e^{-\alpha s^{1/2}}/s^2 \leftrightarrow t(1 + \alpha^2/2t) \operatorname{erfc}(\alpha/2t^{1/2}) \\ - (2t/\pi^{1/2})(\alpha/2t^{1/2}) e^{-\alpha^2/4t} \quad (\text{A.85d})\end{aligned}$$

A.13. LINEAR SYSTEMS

The term *system*, as used here, refers to a set of objects and/or concepts such that an input signal applied to the system results in an output signal. We concern ourselves solely with the relation between the input and output signals, and not with the inner workings of the system.

A system is *linear* if the output $h(t)$ is proportional to the input signal of $g(t)$. If we were to study the inner workings of the system, we could, in principle, describe the relation between $h(t)$ and $g(t)$ by a linear differential equation with constant coefficients. Because the order of the equation is rarely more than 2, we take as a typical equation

$$\frac{d^2 h(t)}{dt^2} + a_1 \frac{dh(t)}{dt} + a_2 h(t) = g(t) \quad (\text{A.86})$$

Taking Fourier transforms and using Equation (A.61a), we get

$$(j\omega)^2 H(\omega) + a_1(j\omega) H(\omega) + a_2 H(\omega) = G(\omega)$$

so

$$H(\omega) = F(\omega) G(\omega)$$

where

$$F(\omega) = \{ (j\omega)^2 + (j\omega a_1) + a_2 \}^{-1} \quad (\text{A.87a})$$

$F(\omega)$ being the *transfer function*. If we know $F(\omega)$, we can find $H(\omega)$ for any input $G(\omega)$. If we apply a unit impulse $\delta(t)$ to the input, then $G(\omega) = 1$ from Equation (A.63) and the transform of $F(\omega)$ gives the output, that is,

$$F(\omega) \leftrightarrow f(t) = \text{unit impulse response}$$

Thus, application of a unit impulse at the input of a

linear system gives us the data to calculate the output for any other input.

Equation (A.87a) can be written in terms of Laplace transforms, the only difference being that use of Equation (A.78a) adds terms in $h(0+)$, $h'(0+)$, and $h''(0+)$ to $G(s)$ [which replaces $G(\omega)$ in Eq. (A.87a)]; however, the usual assumption is that the system is *initially relaxed*, meaning that $h(0+) = 0 = h'(0+) = h''(0+)$, so that Equation (A.87a) becomes

$$\left. \begin{array}{l} \text{where} \\ H(s) = F(s)G(s) \\ F(s) = (s^2 + a_1s + a_2)^{-1} \end{array} \right\} \quad (\text{A.87b})$$

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